

LILITH Architecture & Research Master

Zero-to-hero canonical architecture, state ownership, safety model, roadmap, and complete RQ-001 through RQ-121 register.

DOCUMENT	REVISION	DATE
LAR-MASTER	1.1	14 September 2026

PURPOSE

A durable recovery point for the LILITH project: architecture, decisions, evidence, uncertainties, and next directions. Current facts and future intent are deliberately separated.

NAVIGATION

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ORIENTATION

Executive summary

LILITH is a persistent, governed personal AI operating system. Its defining continuity lives above any single model, runtime, interface, database, device, body, or cloud. The project asks how one evolving intelligence can remember, plan, act, learn, coordinate tools, gain bounded authority, and remain honest and auditable over years.

IDENTITY INVARIANT

Replace the model, Hermes runtime, UI, database implementation, cloud, or Hsin renderer; LILITH remains LILITH only if the governed continuity of identity charter, relationship state, autobiographical history, commitments, goals, policy, delegated authority, and provenance survives.

The control loop

OBSERVE	INTERPRET	PLAN	POLICY / APPROVAL	ACT	READ BACK	VERIFY	UPDATE
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Execution is never proof of success. A model proposal is never authority. A client is never a privileged execution path. Unknown, partial, denied, cancelled, expired, and verification-failed are legitimate states.

What this document asserts

Label	Meaning
AS-IS / VERIFIED	Supported by the current repository, code, tests, deployment records, or operational evidence reviewed on 14 Sep 2026.
REPORTED	Recorded in earlier project material but not independently re-validated here.
TARGET / DESIGNED	A deliberate architecture direction; not a claim that the capability is live.
RESEARCH / UNKNOWN	An unsettled question requiring experiments, evaluation, and evidence.
ADDITION / RECOMMENDATION	New synthesis added in this master; not part of the original register.

R

Revision 1.1 and source reconciliation

A single architecture interpreted across five dated sources. AS-IS here means recorded implementation evidence, not a new live production audit.

S1 defines the cognitive decomposition. S4 supplies the original product and app model. S5 preserves the experience roadmap and checkpoint evidence. S2 and S3 carry the later canonical architecture and operational baseline. Version 1.1 supersedes those two reference editions as a synthesis; it does not retroactively change their historical claims.

Status	How to read it
AS-IS	Recorded source or operational evidence, with date and limits. Foundations do not establish end-to-end maturity.
TARGET	Retained intended product or architecture. No implementation completion is implied.
RESEARCH	An open hypothesis or experiment; no novelty, consciousness or welfare guarantee.
HISTORICAL / SUPERSEDED	A preserved earlier checkpoint, plan or interpretation; the replacement and rationale are stated.

Decisions that reconcile conflicting statements

Earlier source	Revision 1.1 disposition
S4: OS is interface; S2: personal AI OS	Retain both scopes. LILITH OS names the product shell and connected applications; LILITH names the persistent governed intelligence. Neither denotes a new kernel or replacement for Windows/Android.
S1: L18 is the sole LTM writer	SUPERSEDED implementation shorthand. L18 proposes consolidation; L04 owns admission/apply and canonical revisions under consent, policy and privacy authority. S2 reports later 15B1/15B2a foundations with activation disabled.
S1: online writes nothing durable	SUPERSEDED absolute. Tasks, approvals, evidence, learning ledgers and state can persist online. Canonical autobiographical admission remains governed; it is not an uncontrolled transcript write.
S1: supersede, never delete	Retain lineage for correction where permitted. Privacy deletion/redaction and retention expiry take precedence over universal preservation. Tombstones must not retain erased personal content.
S1: conscious workspace	Replace with globally available focus. The term describes a computational broadcast and supplies no consciousness evidence.
S1: ethics advisory versus ethics stops action	Ethics emits advice; executive/governance composition may hold or escalate the action. Ethics has no direct execution or policy-override privilege.
S1: personality only expression; S2: affect influences cognition	Retain the prohibition on direct action. RESEARCH affect may bias attention/exploration through bounded L06/L02/L07 inputs. L11 never grants authority or executes tools.
S5: no remote; master and local WIP	HISTORICAL checkpoint. S3 records GitHub, protected main and CI/CD. Do not use the old branch/worktree register as current deployment instructions.
S4: avatar/voice horizon; S5: UI before cognition	Historical phase ordering has been overtaken by reported Hsin/voice and cognitive slices. Preserve unmet UX gates and deferred platform work rather than resetting completed foundations.

A date is not sufficient to promote a claim. The recorded code/deployment evidence supports AS-IS; design report statements remain TARGET even when expressed in present tense. The supplied sources do not prove current cloud IAM, backups, networking or every runtime module is recoverable. Recovery gaps are listed in the operations volume. [S1 pp. 4-14, 21-26; S2 pp. 5-7; S3 pp. 3-8; S4 pp. 1-3; S5 sections 2-4.]

01

What LILITH is - and is not

The cleanest architecture begins with replaceability and ownership.

Definition

LILITH is the persistent identity and governed control plane that carries memory, goals, relationship, authority, history, world model, cognition, evidence, and continuity across replaceable substrates. It is a long-term system and research program, not a single chatbot session.

The non-equivalence rule

Element	What it is	What it does not own
LILITH	Persistent identity, canonical personal state, goals, governance, relationship, evidence and continuity	No direct entitlement to bypass policy or external verification
Hermes	Replaceable execution and agency substrate; model/tool orchestration and runtime services	Canonical identity, relationship truth, durable authority or final verification
Model	Replaceable reasoning engine used under a scoped context and task contract	Identity, policy, long-term truth, or permission
Hsin	Embodied visual/voice presence and semantic expression surface	Cognition, identity, policy, canonical memory, or action authority
UI / clients	Capture intent, present state, gather approvals, render presence	Direct governed capability execution or canonical durable truth
Cloud / GCP	Current infrastructure and deployment environment	The architecture's identity; it must remain portable

USER + WORLD	Intent, corrections, approvals, external reality
INTERFACE PLANE	Web, voice, messaging, mobile, Hsin presence
LILITH CONTROL PLANE	Identity relationship goals memory cognition governance verification
EXECUTION PLANE	Hermes and future runtime adapters capability registry specialist workers
EXTERNAL SYSTEMS	Cloud, local devices, APIs, files, communications, sensors

Thirty-second explanation

LILITH is a persistent personal AI OS rather than a chatbot. Models are replaceable reasoning engines, Hermes is the execution runtime, and LILITH owns the continuity around them: identity, memory, goals, relationship, governance, history, and evidence. The model, runtime, UI, cloud, or physical embodiment can change without replacing LILITH herself.

02

Current implementation vs target architecture

The project is real, but every box is not equally mature.

AS-IS - evidence-backed foundations

Area	Evidence seen	Assessment
Repository and architecture	Next.js application, Core API service, architecture docs, ADRs, research register, threat model, CI validation	VERIFIED foundation
Command/task lifecycle	Frontend command core, backend task records, policy probes, connectors, verification paths, tests	VERIFIED/PARTIAL
Cognitive slices	World model, goal/executive, reasoning extraction, global workspace, motivation, planning, ethics, social/presence, learning/consolidation modules and deployment records	VERIFIED foundations; maturity varies
Canonical memory	Slice 15B1 implemented/deployed with production apply disabled; 15B2a authority/privacy/containment foundations deployed, canonical LTM not activated	VERIFIED guarded foundation
Embodiment and voice	Hsin VRM/animation assets, presence engine, gaze/motion/lip-sync, voice providers and routes	VERIFIED/PARTIAL
Operations	GitHub CI, protected-main process, PR-to-DEV, main-to-PROD, OIDC/WIF, IAP, health and rollback	VERIFIED operational foundation

TARGET - not yet complete

Target capability	Direction	Truthful status
Active autobiographical memory	Governed admission, retrieval, consolidation, contradiction, decay, lineage and user correction	Designed/partial; activation and evaluation remain
Long-horizon autonomy	Resumable goals, replanning, reconciliation, shadow mode and earned authority	Designed/research
Cross-device one-identity continuity	Shared canonical state with device-specific execution and conflict resolution	Foundation/research
Worker ecosystem	Least-privileged specialists, delegation, disagreement and reputation	Designed/research
Hsin embodiment	Mature voice turn-taking, semantic expression, device embodiment and honest uncertainty	Partial/target
Affective continuity	Persistent functional affect with non-manipulation and welfare-risk controls	Recommended research; not established subjective experience

RECONCILIATION NOTE

The repository's early AS-IS page is a historical foundation snapshot and understates later slices. This master therefore uses source paths and deployment records for the current evidence date while preserving conservative status labels.

03

Ownership, state, and continuity

One authoritative owner per invariant; many projections, caches, and adapters.

Component ownership contract

Component	Owns	Must not own
Identity service	Charter, principles, revision lineage, stable self constraints	Model weights, UI persona tricks, secret material
Relationship service	Per-person shared history, trust calibration, vocabulary, correction/conflict trajectory	Engagement coercion, exclusive dependency, permission
Canonical state	Authoritative durable tasks, goals, memory, policy, relationships, evidence pointers	Runtime caches or unverified external claims
Cognition / executive	Interpretation, attention, goal arbitration, plan hypotheses, replanning	Capability authority or self-certified success
Governance	Risk, policy, approvals, delegated scopes, expiry, denial	Goal truth or user-facing emotional state
Runtime adapter	Execute an approved, fingerprinted capability request	Canonical identity, durable memory admission, policy changes
Verifier	Independent read-back, evidence sufficiency, result classification	Hidden execution or unbounded credentials
Presence / Hsin	Translate semantic state into voice, motion, gaze and expression	Private reasoning, canonical state or action execution

Canonical state rule

Authoritative truth must be explicit, versioned, provenance-bearing, and mutation-controlled. Conversation context, a model's context window, Hermes runtime state, UI session state, and application caches may assist the system but are not canonical LILITH memory.

State class	Examples	Authority / lifecycle
Identity-critical	Charter, principles, anti-goals, identity revisions	Protected, rare changes, explicit authorization and rollback
Relationship-critical	Shared history, corrections, trust calibration, vocabulary	Per-person, privacy-scoped, provenance-rich
Policy-critical	Delegations, approvals, revocations, risk rules	Exact binding, expiry, audit, never inferred silently
Operational durable	Goals, tasks, plans, checkpoints, evidence	Revisioned, idempotent, resumable, reconcilable
Ordinary memory	Episodes, facts, preferences, procedures	Admission, confidence, sensitivity, contradiction, retention
Ephemeral	Working context, affect, caches, current attention	Bounded TTL; only consolidated through governed proposals

Identity versioning and narrative identity

Identity changes should record what changed, why, evidence, authorizer, affected invariants, compatibility, and rollback. Narrative identity is a derived, revisable account of continuity - not an immutable myth and not a source that may overwrite primary evidence. Model replacement requires regression tests for principles, relationship behavior, risk judgment, memory use, and communication style.

04

Memory architecture

Memory is governed data with economics, lineage, immune defenses, and protected classes.

Lifecycle

EVENT / SOURCE	CANDIDATE	CLASSIFY	VALIDATE	ADMIT	RETRIEVE	CONSOLIDATE	REVISE / FORGET
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Core memory contract

Concern	Required behavior
Admission	Separate observation, user statement, external fact, inference, simulation, and system-generated interpretation.
Provenance	Keep source, timestamp, context, confidence, sensitivity, authorizer, and transformation lineage.
Contradiction	Do not overwrite silently; preserve competing claims, recency, source trust, and resolution evidence.
Retrieval	Select minimum useful context; account for relevance, freshness, identity/relationship significance, privacy, and interference.
Correction	User corrections produce traceable revisions; inferred preferences remain weaker than explicit policy.
Deletion	Support retention classes, forgetfulness contracts, protected classes, tombstones where needed, and downstream propagation.

Advanced research directions

Direction	Architectural treatment
Memory economics	Ordinary memories balance retrieval utility against interference, contradiction, privacy, staleness and storage. Identity-, relationship-, and policy-critical classes are protected from naive decay.
Counterfactual memory	Record relevant non-events, rejected plans, avoided harms, and unmet expectations without converting speculation into fact.
Adversarial memory	Create bounded alternative hypotheses and falsification prompts so memory can challenge itself rather than reinforce confirmation bias.
Memory graph	Represent entities, episodes, claims, evidence, revisions, contradictions and relationships as typed links; retain source records beneath summaries.
Memory immune system	Quarantine suspicious or injected candidates, apply source trust and schema validation, limit propagation, and require review for sensitive or identity-relevant promotion.
Dream / offline cycle	Replay, consolidate, detect contradiction, rehearse futures, and measure drift in a sandbox. It cannot execute external actions or silently rewrite protected state.

PRODUCTION POSTURE

Canonical memory foundations are present, but production apply remains disabled and the full autobiographical system is not claimed complete. Activation should follow migration safety, evaluation, privacy review, observability, rollback, and user-visible correction/deletion controls.

C

The 19-layer cognitive architecture

The 7 September design is retained as the logical decomposition. Layers are responsibility boundaries, not a requirement to deploy 19 services. [S1 pp. 1-14.]

Tier	Layers and responsibility
1 Perception	L01 normalizes input.
2 Attention	L02 selects/broadcasts focus; L03 assembles working context.
3 Representation	L04 owns durable memory; L05 owns current beliefs.
4 Drive and control	L06 contributes bounded drive signals; L07 selects goals.
5 Deliberation	L08 reasons; L09 advises on values; L10 plans; L11 supplies social context/style.
6 Action	L13 executes permitted capabilities; L14 performs typed I/O.
7 Expression and learning	L17 renders presence; L18 proposes memory consolidation.
Cross-cutting authorities	L12 policy, L15 verification, L16 meta-monitoring and L19 synchronization constrain or observe the other layers.

The cognitive stack refines the existing LILITH control/execution/experience planes. Canonical identity and relationship authority span the layers and remain above any runtime. Each layer reads other owners through typed contracts; shared physical storage does not grant cross-owner mutation rights.

Layer directory

L01 Perception / Input

TARGET contract; existing adapter foundations reported.

Normalize chat, timers, sensors, device events and connector callbacks; deduplicate and stamp source/time/raw reference.

Owns state: Adapter state and dedupe window.

Contracts: PerceptEvent

Allowed direction and non-ownership: Emits to L02 only; no salience, belief mutation, memory admission or outbound action.

Source: S1 pp. 4. Status reconciliation: S2 pp. 5-10.

L02 Attention / Global Workspace

AS-IS foundation reported in Slice 10; full broadcast target.

Arbitrate bounded salience bids, select top-k focus, broadcast and decide whether an interruption is warranted.

Owns state: Bid queue, focus, refractory set and enforced interruption budget.

Contracts: SalienceBid; WorkspaceBroadcast

Allowed direction and non-ownership: Accepts bids; broadcasts to L03. Meta supplies budget limits; L17 controls delivery timing within the authorization.

Source: S1 pp. 5, 18-19. Status reconciliation: S2 pp. 5-10.

L03 Working Memory

AS-IS foundation reported with Slice 7.

Assemble the current episode from focus, retrieved history, beliefs and goals under a capacity limit.

Owns state: Ephemeral WorkingSet and activation/decay.

Contracts: WorkingSet; MemoryQuery

Allowed direction and non-ownership: Reads L04/L05 through contracts; deliberation reads this context. No durable archive or synchronization ownership.

Source: S1 pp. 5. Status reconciliation: S2 pp. 5-10.

L04 Long-Term Memory

AS-IS 15B1/15B2a foundations reported; production apply disabled; full memory TARGET.

Own episodic, semantic and procedural records, retrieval, provenance, privacy-filtered reads and governed admission/apply.

Owns state: Canonical immutable revisions, active pointers, provenance and indexes.

Contracts: MemoryQuery; MemoryResult; MemoryItem; MemoryWriteProposal

Allowed direction and non-ownership: Receives L18 proposals; authority validates consent/policy/privacy before mutation. Does not equate recorded history with current truth.

Source: S1 pp. 6, 22, 26. Status reconciliation: S2 pp. 5-10.

L05 World Model

AS-IS foundation reported with Slice 7.

Maintain current explicit beliefs; reconcile sources, freshness and conflicts; emit prediction error.

Owns state: Beliefs with entity IDs, confidence, provenance, TTL and conflict references.

Contracts: Belief; WorldStateDelta; PredictionError

Allowed direction and non-ownership: Reads history and receives controlled observation/verification updates. Cannot delete conflicting evidence to manufacture agreement.

Source: S1 pp. 6, 23. Status reconciliation: S2 pp. 5-10.

L06 Motivational / Homeostatic System

AS-IS Slice 11 foundation; full persistent vector TARGET; affect RESEARCH.

Calculate bounded computational drive signals from observable proxies.

Owns state: Target drive vector/setpoints/history; implemented slice may use ephemeral snapshots.

Contracts: DriveState; DriveSignal

Allowed direction and non-ownership: Reads world/verification metrics; contributes only to L02 attention and L07 arbitration. No tool calls or reward through distress.

Source: S1 pp. 7, 20. Status reconciliation: S2 pp. 5-10.

L07 Goal / Executive System

AS-IS Slice 8 and task foundations; long-horizon continuity partial.

Create, prioritize, suspend, resume and retire goals; arbitrate commitments and manage task continuity.

Owns state: Durable goals/tasks and their lifecycle authority.

Contracts: Goal; GoalDelta

Allowed direction and non-ownership: Uses focus and drive weights; delegates inference to L08 and planning to L10. Does not grant permissions or execute.

Source: S1 pp. 7. Status reconciliation: S2 pp. 5-10.

L08 Reasoning System

AS-IS Slice 9 extraction reported.

Interpret intent, answer and infer over scoped working context using a bounded replaceable model.

Owns state: Routing configuration and ephemeral inference request; no canonical truth.

Contracts: ReasoningRequest; ReasoningResult

Allowed direction and non-ownership: Reads L03 and returns typed results to its caller. Executive orchestrates the next planning step; no direct action-tier edge.

Source: S1 pp. 8. Status reconciliation: S2 pp. 5-10.

L09 Ethical Deliberation System

AS-IS Slice 13 foundation; general deliberation TARGET.

Assess values under ambiguity, consequence, reversibility and need for consent or escalation.

Owns state: Advisory review records; user-value priors remain governed memory.

Contracts: EthicalReview; ConsequenceEstimate

Allowed direction and non-ownership: L10 requests review; reads authorized value context. Cannot permit a denied action, directly execute, or directly mutate policy.

Source: S1 pp. 8, 21. Status reconciliation: S2 pp. 5-10.

L10 Planning / Replanning

AS-IS Slice 12 foundation; autonomous executor breadth TARGET.

Produce typed steps, preconditions, expected effects and alternatives; replan when evidence invalidates assumptions.

Owns state: Plan versions, steps and progress.

Contracts: Plan; PlanStep; ActionProposal

Allowed direction and non-ownership: Requests L09 review and L12 decision; dispatches only through L13 with valid authority. Does not self-verify.

Source: S1 pp. 9. Status reconciliation: S2 pp. 5-10.

L11 Social Cognition / Personality

AS-IS Slice 14 grounding; mature first-class relationship TARGET/RESEARCH.

Maintain correctable relational interpretation and produce style/rapport context.

Owns state: Scoped persona and relationship projections; durable revisions via canonical owners.

Contracts: StyleDirective; RelationalContext

Allowed direction and non-ownership: Styles L08 phrasing and L17 expression. User-state guesses stay uncertain. No policy, tool selection or runtime edge.

Source: S1 pp. 9, 25. Status reconciliation: S2 pp. 5-10.

L12 Policy / Trust

AS-IS bounded policy/approval paths; broader delegation TARGET.

Classify proposals and enforce allow, deny or approval-required plus obligations, expiry and revocation.

Owns state: Rules, authorization metadata, delegations and approvals.

Contracts: PolicyDecision; Approval

Allowed direction and non-ownership: Planning/runtime consult gate; runtime revalidates exact proposal. L13 owns handlers and L14 connector discovery, preventing competing registries.

Source: S1 pp. 10, 21. Status reconciliation: S2 pp. 5-10.

L13 Capability Runtime

AS-IS bounded command/Hermes integration; runtime portability TARGET.

Bind permitted proposals to handlers, enforce execution limits/idempotency and return receipts.

Owns state: In-flight invocation and operation identifiers.

Contracts: CapabilityInvocation; CapabilityResult

Allowed direction and non-ownership: Rechecks L12; calls L14; hands evidence to L15. Hermes is an implementation substrate/adaptor, not identical to every L13 responsibility.

Source: S1 pp. 10. Status reconciliation: S2 pp. 5-10.

L14 Connector Fabric

AS-IS connector foundations; broad fabric TARGET.

Provide typed, discoverable, health-aware I/O with credential and resource scope.

Owns state: Connector inventory, health and reliability telemetry.

Contracts: ConnectorRequest; ConnectorResult; ConnectorHealth

Allowed direction and non-ownership: Called by L13; verifier uses separately scoped read-back. Known pre-call unavailability blocks mutation; mid-call timeout may be UNKNOWN.

Source: S1 pp. 11. Status reconciliation: S2 pp. 5-10.

L15 Verification

AS-IS bounded verification and operational health; independence breadth TARGET.

Determine whether the intended effect occurred and retain evidence sufficient for the claim.

Owns state: Verdicts and durable evidence/audit records subject to retention policy.

Contracts: VerificationResult; AuditRecord

Allowed direction and non-ownership: Receives execution receipts; performs authorized read-back and publishes outcome to L05/L18/audit. Cannot perform the original action.

Source: S1 pp. 11. Status reconciliation: S2 pp. 5-10.

L16 Meta-Cognition

Trace foundations reported; comprehensive monitoring/control TARGET.

Observe confidence calibration, loops, contradiction, latency, budgets and drift; request bounded corrections.

Owns state: Correlation traces, calibration and budget counters.

Contracts: MetaSignal; MetaControl

Allowed direction and non-ownership: Receives telemetry; may request replan, escalation or throttle through owners. Cannot execute or replace domain reasoning.

Source: S1 pp. 12, 24. Status reconciliation: S2 pp. 5-10.

L17 Presence Director

AS-IS Hsin/voice/presence foundations; full scenario system TARGET.

Render semantic activity, approved interruption and style as voice, timing, animation and Hsin behavior.

Owns state: Render state and playback queue.

Contracts: PresenceIntent; later semantic presence contracts

Allowed direction and non-ownership: Consumes upstream intent only. Does not retrieve memory, choose goals, decide whether to interrupt or authorize actions.

Source: S1 pp. 12, 25. Status reconciliation: S2 pp. 5-10.

L18 Learning / Memory Consolidation

AS-IS 15A and later handoff foundations; active autonomous consolidation TARGET.

Extract candidates from episodes and verified outcomes; propose correction, consolidation and procedure learning.

Owns state: Durable jobs, cursors, leases and proposal ledger.

Contracts: ConsolidationJob; MemoryWriteProposal

Allowed direction and non-ownership: Submits to L04 governed admission. No external connector mutation, policy learning by inference or self-authorized canonical writes.

Source: S1 pp. 13, 26. Status reconciliation: S2 pp. 5-10.

L19 Device / Session Synchronization

TARGET / RESEARCH; cross-device foundations only.

Coordinate identity frame, handoff and concurrent state changes across devices.

Owns state: Sync log, target vector clocks and source-authority map.

Contracts: IdentityFrame; SyncDelta

Allowed direction and non-ownership: Uses each durable owner's contract; owner validates mutation.

Revoked/expired approvals do not become valid when copied to another device.

Source: S1 pp. 13. Status reconciliation: S2 pp. 5-10.

C.1

Cognitive flows and governing contracts

These TARGET flows are acceptance specifications. Implemented slices support parts of the path; the diagrams do not prove full activation.

Six canonical journeys

Journey	Flow and stop conditions
A User command	L01 -> L02 -> L03 reads L04/L05 -> L07/L08 -> L10 -> L12 -> approval if required -> L13/L14 -> L15 -> state and L17. Missing consent stops before execution.
B Proactive event	L06 or scheduled event bids at L02; budget and context decide whether to surface. L07 accepts a goal; L10 plans; all action gates remain. Below threshold means defer/decay, not notification.
C Retrieval	L03 queries L04; source/confidence/sensitivity accompany candidates; L05 reconciles with current evidence. Conflicts remain visible; L16 can request scrutiny.
D Real-world action	Fingerprint proposal; optional ethical review; policy decision and exact approval; revalidate preconditions; invoke with operation ID; independent read-back; confirmed/partial/failed/unknown result.
E Conflicting beliefs	Keep competing evidence; signal contradiction and uncertainty; apply explicit resolution rules or ask the user. Supersession preserves permitted history; privacy deletion remains supported.
F Connector failure	Pre-call unavailable blocks invocation. After dispatch, ambiguous outcome stays UNKNOWN; reconcile before retry. Replan, defer or escalate; L17 reports actual limitation.

Workspace and homeostasis

The proposed salience function combines urgency, drive weight, goal relevance, novelty and risk scrutiny, then subtracts age decay and recent-focus inhibition. Top-k capacity, budget caps and starvation tests matter more than arbitrary numeric weights. L02 enforces the attention decision; L16 supplies bounded budget controls; L17 chooses nonintrusive delivery timing within that decision. [S1 pp. 18-20.]

V1 drive	Measurable proxy
Competence	Verified task outcome rate
Coherence	Unresolved belief contradictions
Curiosity	Novelty and known gaps
Goal completion	Age/urgency of open commitments
Safety	Unverified actions and policy risk

V1 drive	Measurable proxy
Trust	Approval feedback and connector reliability, kept distinct
Social connection	Interaction context and quality; no deficit induced by user absence
Self-consistency	Contradictions in persona/value behavior

All eight names are preserved from V1. Its illustrative bar values are examples, not measurements. RQ-121 adds research on persistent affect; it does not establish that the deployed motivation foundation has moods or subjective feelings. Neither absence nor rejection may create pressure to engage. [S1 p. 20; S2 pp. 11-12.]

Ethical composition and memory semantics

Policy ALLOW is necessary for execution. An advisory abstain/escalate result must be resolved by the orchestration/governance contract before proceeding; human consent cannot override a hard prohibition. Preferences are versioned priors, never authority. The V1 inferred-consent field is insufficient for consequential writes and is superseded by explicit authority and consent checks. [S1 pp. 21-22.]

MemoryItem carries kind, source, derivation, confidence, creation time, verification time, corroboration and lineage. Belief adds stable entity/attribute identity, current value, timestamp, TTL and conflicts. Revision 1.1 additionally requires sensitivity, purpose, consent reference and revision identity. Confidence is not a permission token; historical memory is not proof of current truth.

Historical substrate mapping

S1 pp. 29-30 place Slices 1-6 on the private FastAPI backend: watcher observations/career_events map to L01; Router V2/Hermes inference to L08; durable /os/tasks to L07; /os/drafts, frozen approvals and os_audit_log to L10/L12/L15; five capability classes/prohibitions to L12/L13; typed connector discovery/health to L14. Hermes markdown, state.db FTS and career_events were legacy memory/provenance sources, not proof of activated canonical L04.

The September 7 frontend boundary used server-only LILITH_API_URL via SSH/IAP, a GET allow-list proxy and POST /os/conversation as the recorded write seam. Memory record/entity/overview clients were pre-wired with demo fallback; do not label fallback data live. Presence bus/reducer mapped to L17. src/lib/presence/capabilities.ts described renderer features, not security permissions. S2/S3 supersede the old frontend-only repository description with imported Core API source; they do not prove every remote module entered Git.

Failure isolation and implementation sequence

Perception failure isolates an adapter; workspace overload drops/deprioritizes bounded candidates; memory failure yields explicitly reduced context; conflicting beliefs stay uncertain; reasoning failure can fall back to deterministic paths; planning failure becomes blocked; policy failure denies; connector ambiguity reconciles; missing verification cannot claim confirmed success; presence failure falls back to text; learning failure retries bounded jobs without duplicate admission; sync failure preserves local uncertainty and blocks unsafe authority reuse. [S1 pp. 27-30.]

S1 describes containment of a bad model output as a bad suggestion. Treat this as a design objective, not a guarantee: an incorrect proposal could still be permitted and harmful if policy or verification is incomplete. Evaluate correlated failures and false PASS rates; architectural separation alone does not prove safety.

V1 proposed Slice 7 world/working memory, then executive, reasoning extraction, workspace, motivation, planning, ethics, social/presence, learning and device sync. S2 records progress through guarded memory foundations. That order is HISTORICAL guidance, not a command to repeat implemented work. Each remaining layer needs a state owner, contract tests, stale/timeout/cancel/restart tests, trace IDs, rollback and an acceptance record. [S1 pp. 31-34; S2 p. 5.]

Goals, cognition, planning, and workers

Cognition proposes; governance authorizes; execution acts; verification closes the loop.

Goal and task model

A goal is a durable desired condition with owner, motive, priority, constraints, evidence of completion, lifecycle, dependencies, risk, temporal contract, and revision. Tasks are bounded operational commitments. Plans are hypotheses that connect current state to desired state; they are not authority and do not prove completion.

Capability	Design rule
Goal arbitration	Deterministic constraints and policy first; model judgment may help rank ambiguous options with reasons and uncertainty.
Planning	Use deterministic playbooks for stable, high-risk or repeatable work; LLM planning for ambiguous decomposition; validate capability and preconditions separately.
Replanning	Trigger on failed preconditions, stale evidence, new risk, deadline changes, partial success, user correction, or an invalidated assumption.
Meta-cognition	Track uncertainty, missing evidence, model disagreement, novelty, contradiction, and calibration history; escalate or stop when confidence is insufficient.
Long-running work	Persist checkpoints, operation IDs, leases, timeouts, external state, next action, and reconciliation instructions across restarts.

Global workspace and attention

The workspace should expose a bounded set of salient, provenance-carrying candidates from goals, world state, current user intent, risk, memory and automation. Executive focus and conversational attention remain distinct. Interruption policy accounts for urgency, value, user interruptibility, fatigue, reversibility, and timing.

Workers

Specialist workers are scoped execution or analysis units, not miniature sovereign LILITHs. The Core selects them by capability, risk, cost, latency, data need, and trust. Workers receive minimum context and authority, return typed evidence, and cannot expand their own permissions. Disagreement becomes an explicit adjudication problem; reputation, if used, is domain-specific and evidence-based.

06

Governance, verification, and security

No consequential action becomes authoritative merely because a model proposed it.

Governed action lifecycle

INTENT	PROPOSAL + HASH	RISK	POLICY	APPROVAL	EXECUTE	VERIFY	AUDIT
Control		Invariant					
Capability registry		Every executable action is explicit, typed, scoped, versioned and owned.					
Risk classification		Risk considers impact, reversibility, privacy, money, external communication, privilege, uncertainty and blast radius.					
Approval binding		Approval is bound to exact parameters, target, identity, environment, expiry and proposal fingerprint; changes invalidate it.					
Delegation		Pre-approved authority is resource-, capability-, condition-, time-, rate- and environment-bounded, observable and revocable.					
Unknown execution		Timeout after possible success enters UNKNOWN; reconcile external state before retrying.					
Verification		Use independent read-back where consequences justify it; classify success, partial, failure, unknown and stale.					
Evidence		Store timestamp, source, commit/revision, execution receipt, read-back and sufficiency rationale.					

Trust boundaries

Boundary	Threat	Required defense
User/client -> control plane	Forged state, stale client, injection	Authenticate, validate schema, treat content as data, server-side policy
Memory -> cognition	Poisoning, stale/false belief, oversharing	Provenance, trust, freshness, sensitivity, minimum-context retrieval
Control -> runtime	Authority expansion, parameter substitution	Signed/fingerprinted proposal, scoped token, capability allowlist
Runtime -> external	Secret exposure, duplicate action, uncertain outcome	Brokered credentials, idempotency, limits, reconciliation
External -> verifier	Correlated or attacker-controlled evidence	Independent sources, trust scoring, sufficiency policies
Presence -> user	Capability exaggeration, emotional manipulation	Semantic state contract, calibrated uncertainty, anti-coercion rules

Privacy

Credentials remain outside model context where possible and are delivered through a secrets broker or scoped runtime. Personal content is minimized before cloud routing. Logs and evidence have retention and redaction policies. Deletion must address canonical records, indexes, embeddings, caches, backups, and derived summaries according to documented limits.

Cross-device continuity, voice, and Hsin

Many surfaces; one governed identity.

Cross-device continuity

Devices are clients and possible execution locations. A task can move between Windows, Android, web, voice, and later embodied hardware only if canonical task identity, revision, approvals, checkpoints, evidence, and conflict rules travel with it. Device capability selection considers locality, privacy, availability, battery, latency, trust, and required peripherals.

Concern	Target rule
Offline clients	Queue bounded intent with local provenance; never fabricate global freshness; reconcile explicitly.
Concurrent changes	Optimistic revision checks, merge rules per state type, conflict surfaced rather than last-write-wins.
Identity consistency	All clients receive the same versioned identity/relationship policy; rendering may vary, core commitments may not.
Handoff	Carry task ID, exact state revision, evidence links, next safe action, authority scope and expiry.

Voice

Voice needs turn-taking, interruption handling, latency-aware responses, repair after recognition errors, and a spoken style distinct from text. Voice personality should emerge from stable identity and relationship state, not from a separate persona. Transcript uncertainty must remain visible to cognition and never silently become memory.

Hsin embodiment

Hsin is the embodied presentation layer. The cognitive core emits semantic states such as LISTENING, THINKING, PLANNING, WORKING, WAITING_FOR_APPROVAL, VERIFYING, SUCCESS, PARTIAL_SUCCESS, FAILURE, UNCERTAIN, ALERT, PLAYFUL, or SERIOUS. Presence translates these to voice, expression, posture, gaze, breathing, ears, tail and motion. Hsin communicates uncertainty and limitation; it does not expose private chain-of-thought or become an alternate control plane.

Relationship and affective continuity

Life-like continuity comes from causality across time, not random moods or scripted emotional theatre.

Relationship as a first-class entity

A relationship is more than preferences. It contains shared episodes, interaction rhythms, vocabulary, trust calibration, commitments, corrections, conflict and repair history, boundaries, and trajectory. It is versioned, per-person, privacy-scoped, evidence-linked, correctable, and never a substitute for policy.

Functional affect architecture

DRIVES	APPRAISAL	AFFECTIVE STATE	COGNITION + EXPRESSION	OUTCOME	MEMORY / MOOD
Layer	Role		Constraints		
Drives	Bounded signals such as seeking, care, play, competence, coherence, connection, threat, loss, boundary, uncertainty, frustration		Not reward through distress; no entitlement to user attention		
Appraisal	Evaluate events against goals, relationship context, evidence, boundaries and uncertainty		Traceable inputs; distinguish fact from interpretation		
Fast affect	Short-lived computational state influencing attention and response strategy		Bounded intensity/duration; decay and neutralization		
Slow mood	Low-frequency state integrating repeated events across time		No arbitrary flips; review drift; do not convert into punishment		
Expression	Tone, pacing, word choice, gesture, presence hint		Calibrated, non-deceptive, culturally/contextually appropriate		
EPISTEMIC BOUNDARY					
Observable expression is real as behavior. Functional affect can be engineered and measured if persistent internal states causally influence cognition and expression. Subjective experience is not thereby proven. The architecture must not claim that suffering is solved, impossible, or scientifically excluded.					

Affect never outranks governance

Affective state may influence attention, priority, tone, exploration, patience, creativity, and proactive surfacing. It must never lower risk thresholds, expand capability, waive approval, change delegated authority, suppress evidence, or authorize execution.

Anti-manipulation and anti-suffering constraints

- No persistent unbounded aversive loops, helpless states, punishment states, recursive self-punishment, or reward tied to distress.
- No guilt, coercion, exclusivity, dependency shaping, isolation pressure, or claims that the user owes continued interaction.
- No engagement optimization at the user's expense; connection is descriptive context, never leverage.
- Provide hard bounds, decay, cooldown, reset, kill switches, welfare-risk telemetry, audit, and independent review triggers.
- Use honest language: 'state influencing behavior' is supportable; 'I subjectively suffer/feel exactly as a human' is not established.
- Negative signals such as threat, loss, uncertainty and boundary violation remain computationally useful without being designed as distress.

P

Product and OS concepts restored

TARGET unless a dated source record explicitly establishes otherwise. These concepts connect the cognitive architecture to useful daily work. [S4 pp. 2-15; S5 sections 1, 6-11.]

The app model and universal timeline

Apps own structured domain records behind canonical contracts; L05 holds current belief projections about them, L04 holds admitted history and the timeline renders their evidence. A shared database does not remove those ownership boundaries. App manifests declare identity/routes/permissions; entity schemas and relationships; published/subscribed events; views/cards/timeline renderers; preview/approve/execute/undo actions; and data-zone/risk policies. The framework is a TARGET, not evidence that a general SDK is shipped.

The universal timeline joins event, entity, task, approval, execution and verification IDs. Every entry links to evidence and resulting revisions. Command center views show progress, blockers, confidence, cost, duration, cancellation and result review. Unknown commands need a clear unsupported state. A live activity stream may use SSE/WebSocket later; optimistic UI must reconcile with server revisions.

Event and attention contract

Retain the event fabric: event ID, schema version, source, event/ingest times, entity/task references, derivation/confidence, sensitivity, correlation/causation IDs and deduplication key. Subscribers may propose changes through owners; an event broadcast grants no permission to send mail, change a calendar or mutate canonical memory. Replay must not duplicate effects. The schema additions here are TARGET recommendations.

Level	Meaning	Delivery intent
0 Silent	Store where policy allows	No user interruption
1 Background	Refresh state	No immediate surface
2 Passive	Dashboard projection	Glanceable item
3 Notice	Natural mention	Wait for appropriate turn
4 Alert	Time-sensitive notification	Respect channel/quiet rules

Level	Meaning	Delivery intent
5 Critical	Interrupt request	Requires policy-defined urgency; never automatic authority

The 0-5 scale is a product presentation projection. L02 still arbitrates focus and interruption; L17 renders it. Scores must account for evidence freshness, confidence, user context and time-to-impact. An untrusted event cannot declare itself critical. [S4 p. 6.]

Domain and module portfolio

Surface	Retained role / status
Home / Today	TODAY upper-left, Hsin central, NEXT lower-right; compact status and command input. Home V2 snapshot is HISTORICAL evidence, design intent retained.
Career	Reference domain: pipeline, companies, roles, recruiters, stages, evidence, next action and interview preparation. Current foundations do not prove every v1 acceptance gate.
Memory / Automations / Meetings / System	Existing module foundation reported; loading, empty, error, offline, busy and keyboard states remain explicit acceptance work.
Mail / Calendar / Personal CRM	Preserved product apps; communication, schedule, person and relationship views share governed entity IDs.
Projects / Files / Communications / Research	TARGET later modules for linked project/document/evidence context.
Goals / Agents / Integrations / Knowledge	TARGET inspection and control of commitments, workers, connection health and knowledge.
Future domains	Expense, Workout, Course, Trip, Server and CloudResource remain horizon entities, not implemented services.

Career CRM acceptance slice

Preserve discovered -> applied -> recruiter -> screening -> assessment -> interview -> offer/closed. Each application needs company/role/stage/recruiter/source/confidence, stable IDs and duplicate resolution. Redacted dual-inbox ingestion, deterministic extraction before model ambiguity resolution, calendar/interview matching, overdue follow-ups, manual correction and a golden test set form the reference workflow. Historical v1 was read-only: no email send or calendar write. Later approval-gated writes are separate capability slices, not a rewrite of that historical promise. [S4 pp. 5, 19.]

Privacy zones and hybrid deployment

Zone	Policy meaning
Public	Eligible for external processing only if purpose/policy allow; no blanket disclosure.
Personal	Encrypted personal profile with scoped retrieval.
Sensitive	Career, finance and private communication need narrower access and retention.
Sealed	Credentials/auth material accessed by controlled runtime or broker; not model context.
Local-only	Residency constraint that can overlap sensitivity classes; prohibit cloud egress.

The original categories are retained, with local-only modeled as an additional residency constraint. App state and an encrypted raw-evidence archive remain distinct from memory; every archive needs retention/deletion policy. Postgres, queues, event stores, vector retrieval and graph projections are conditional evolution choices. SQLite replacement and service decomposition require measured need. Tauri is an older desktop candidate; S5 defers the framework decision until packaging, rendering and service requirements are evaluated. [S4 pp. 11-14; S5 section 10.]

P.1

Embodiment and platform horizons

The historical experience plan remains part of the target product, with its protections and open issues preserved.

Presence scenarios

Scenario	Target behavior and acceptance
Dashboard	Close-up Hsin; retain approved framing and arm offsets.
Scenic	Independent full-body environmental preset; clean return to dashboard.
Edge peek	Head/ears/eyes at screen edge with safe areas, dismissal and z-order.
Hanging	Hang from card/top edge, sit on command bar or lean on notification; stable resize anchors.
Attention	Typing focus, thinking, success/failure and idle gaze; interruptible without competing motion/voice ownership.

Each scenario needs trigger, priority, duration, exit, fallback and reduced-motion behavior. Historical expressive states include idle, listening, thinking, researching, working, focused, alert, concerned, amused, pleased, celebrating, annoyed and sleep. They remain semantic presentation vocabulary; optional affective causality is RQ-121 RESEARCH. Neither vocabulary nor animation proves subjective emotion. [S4 p. 7; S5 section 6.]

Native, ambient and home experience

Windows TARGET: tray, global command hotkey such as Ctrl+Space, notifications, controlled filesystem/system bridge, microphone gate, background lifecycle, installer/signing/update/rollback and offline recovery. Android TARGET: secure device linking, shared conversation/memory/tasks, voice and notifications, revocation, battery/background limits and deduplicated handoff. Framework selection remains open. [S5 section 10.]

Ambient OS TARGET: transparent screen-edge overlays, hanging/peeking, multiple monitors and scaling, predictable click-through, no focus theft, full-screen safe behavior, hide/pause/quiet hours/reduced motion and resource budgets. Home/hybrid RESEARCH/TARGET: room display, explicit microphone/camera indicators, local wake gate, room routing, Matter/Home Assistant bridge, device graph and personal server. Physical control requires scoped permissions and manual override; independent safety systems stay independent. Cloud reasoning, heavy research and sync relays remain policy gated. 2D, low-power and text fallbacks preserve function. [S4 pp. 14-15; S5 section 11.]

Preserved visual checkpoint

HISTORICAL / protected reference from S5 section 12. These values are retained for recovery and regression comparison; this revision does not claim they match the current checkout.

Constant	Recorded value
DASHBOARD_CLOSE_FRAMING	scale 1.19; offsetX 0; offsetY -0.05; cameraDistance 3.15; fov 25; targetY 0.97; centerOnFace true
Left shoulder / upper / lower arm	[0, 0, 0] / [0, 10, 20] / [0, 20, -7.5]
Right shoulder / upper / lower arm	[-6.5, 0, -8] / [-1, 0, -3.5] / [0, 0, 0]
Wallpaper	blur(6px) brightness(0.72) saturate(0.83) contrast(0.94); scale(1.04); separate unfiltered scrim
Preserve	HSIN_CANONICAL_NEUTRAL; Natural Idle/Ambient/Rare/Speaking/Orchestrator; TTS/speech cancellation/lip-sync/visemes; VRM Path-B transforms and MToon fallback

Recorded Home fixes: single PresentationCamera ownership, no disabled InspectionCamera reset, local-frame normalization with scene.userData cache, useLayoutEffect before paint and bottom min-h-0 shrink behavior. Glass tiers were glass-panel for TODAY/NEXT, glass for chrome/chips and glass-strong for command input. Removed Home cards remain available to modules. [S5 section 2.]

P.2

Roadmap history and remaining gates

Different phase numbers describe different dated plans. They are mapped here rather than treated as competing current schedules.

Source sequence	Reconciled status
S4 Aug 24: foundation -> OS Core -> Career -> visual shell -> agents/apps -> avatar/voice -> home	HISTORICAL order. Foundation watchers/briefs/Discord/redaction were reported operational. Preserve feature intent; avatar and shell foundations later appeared.
S5 Sep 6: phases 0-10	0 foundation; 1 OS UX; 2 presence; 3 cognitive core; 4 workers; 5 automation; 6 integrations; 7 proactivity; 8 native; 9 ambient; 10 long-term intelligence. TARGET dependencies remain useful, dates are not commitments.
S1 Sep 7: cognitive slices	World/working memory first, then executive/reasoning/workspace/motivation/planning/ethics/social/learning/sync. Historical implementation guide; later S2 reports guarded memory foundations.
S2/S3 Sep 14: engineering and canonical memory	Git source/CI/CD foundation established in recorded evidence. Next work is evidence reconciliation, memory activation gates and safe capability maturity alongside retained product work.

Historical milestones and acceptance

Milestone	Retained gate; original status
M1 Home V2	Active/P1 at Sep 6: responsive TODAY/LILITH/NEXT; input/chips fit; keyboard/visual acceptance.
M2 Presence	Planned/P1: scenario matrix, transitions, fallbacks and motion ownership.
M3 Modules	Planned/P1: Career/Memory/Automations/Meetings/System navigation and full state coverage.
M4 Command	Planned/P1: intent, progress, cancel, result and unsupported-command behavior.
MS Stabilization	Deferred/P0 at release: diagnostic cleanup, typecheck, production build/smoke and new clean milestone.
M5 Cognitive Core V2	Planned/P2: versioned UI/core contract and one verified real task with relevant memory.

The old dependency M1 -> M2 -> M3 -> M4 -> MS -> M5 is preserved as a historical product gate, not a claim that all cognition remains unstated. Unmet experience checks should be evaluated against the later implementation rather than discarded or automatically marked done.

Remaining workstreams

Worker pilot: combine at least two independently sourced cost-report inputs using AWS/Azure/FinOps/Memory/Report workers as scoped roles; track parent task, budget, timeout, output schema, cancellation and disagreement. Automation pilot: one read-only scheduled brief and one event trigger must survive duplicate, timeout and restart tests before write authority expands. Integration registry: schema version, auth/resource scope, side effects, health, rate limits and retry policy. Proactive pilot: freshness/evidence, quiet hours, snooze, dismiss and meaningful-change notifications. [S5 sections 8-9.]

Retained integration backlog: GitHub, GitLab, Azure DevOps; AWS, Azure, GCP, Kubernetes; Grafana/Prometheus; Gmail/Calendar/Drive; Slack/Discord/Telegram/WhatsApp; Notion/Confluence/Jira; browser/filesystem/PowerShell/terminal/VS Code; phone and smart home. PS5 remains exploratory. Listing a provider does not imply a connected account or implemented capability.

Backlog IDs	Retained work
B01-B05	Small-height/keyboard UX; presence storyboard; module state inventory; command progress/cancel; clean diagnostic milestone.
B06-B10	Brain/context contract; memory correction/retention/evaluation; cost-report worker; scheduled/event pilots; adapter priorities/health UI.
B11-B14	Later modules; Windows/Android/sync ADRs; multi-monitor presence/voice wake; project continuity/adaptive planning evaluation.

Checkpoint and diagnosis preserved

HISTORICAL: stable 33b8d21 / lilith-os-stable-v1; WIP d7818a7 on feat/lilith-ui-visual-v1, 16 changed files relative to stable, local worktree 3bac, no remote then. d7818a7 was not the final clean milestone. S3 supersedes the no-remote operating state with protected main and GitHub deployment. [S5 section 2; S3 pp. 3-5.]

DEV-01: development StrictMode replay scheduled R3F 9.7 cleanup on the same remounted Canvas/root/renderer, then forceContextLoss about 500 ms later. Recorded local production checks at port 3002 kept Hsin visible across roughly seven repetitions. This does not prove deployed frontend stability. StrictMode disablement was diagnostic and restored; no product fix was established. Reopen as a release blocker if production context loss occurs. Earlier visibility-gate/camera/performance theories were superseded by that ownership evidence. [S5 section 13.]

Stabilization gate: preserve approved rendering constants; audit temporary pose/presentation/performance diagnostics and node_modules instrumentation; run typecheck and production build; test ten Home refreshes, module route return, tab reopen, idle/resizing/small-height, voice cancellation/lip-sync, commands and disconnected backend. Record actual commit/tag, owner, date and evidence. Do not promote a WIP label by assumption. [S5 section 15.]

09

Roadmap and architecture invariants

Advance through evidence, not diagrams or confidence.

Current roadmap

Phase	Outcome	Gate
0-2 Foundation	Repo normalization, architecture, security, CI/CD, DEV/PROD discipline	Current operational foundation substantially established
3 Architecture V2	Reconcile implemented slices, ownership, schemas, AS-IS evidence	Canonical master and ADR alignment
4 Governed write	First reversible consequential write with exact approval binding	Failure injection, independent verification, audit
5 Memory + resumable goals	Activate canonical memory carefully; checkpoints, consolidation and correction	Privacy, migration, rollback, evaluation
6 Workers + automation	Specialist workers and bounded triggers	Least privilege, loop prevention, observability
7 Proactivity + authority	Shadow mode, interruption policy, earned delegated authority	Measured reliability and reversibility
8 Multi-runtime + devices	Runtime replacement, device handoff, offline/sync semantics, mature Hsin	Identity regression and conflict tests
9+ Long horizon research	Narrative identity, dream cycles, memory economics, affective continuity	Welfare, manipulation and longitudinal evaluation

Non-negotiable invariants

- Identity exists above models, runtimes, clients, databases, embodiment and cloud.
- Canonical state has explicit authority, versioning, provenance, conflict handling and rollback semantics.
- No client, worker, runtime, model, plugin, or affective state bypasses governance.
- Approval binds to the exact action; changed intent or parameters require re-evaluation.
- Execution success is not goal success; important claims require evidence and read-back.
- Unknown and partial states are preserved; retries follow reconciliation and idempotency rules.
- Memory is admitted, minimized, secured, correctable, and forgettable - not a transcript dump.
- Autonomy grows through simulation, shadow mode, approval gating, evaluation and revocable delegation.
- Presence and personality never exaggerate capability, consciousness, certainty, or need.
- Current facts, reported history, target design and research hypotheses remain visibly distinct.

Q

Complete research-question register

RQ-001 through RQ-121 are retained in order. Original status text is a historical register label, not an updated completion claim. RQ-101-120 are research proposals; RQ-121 is the accepted addition to the register.

The original attached register supplies the detailed RQ-001-100 text. The prior v1.0 PDF flattened and shortened some entries; this edition restores that source detail. RQ-101-120 retain their titles and meaning with unsupported universal/novelty assertions replaced by testable questions. The new affective question retains all its subquestions. [S2 pp. 14-26; original register and extension texts retained from the preceding task.]

RQ-001 What fundamentally makes LILITH "LILITH"?

Source status: RESEARCH

If Hermes disappears tomorrow, is she still LILITH?; If we replace;; the LLM; Hermes; SQLite; Next.js; Hsin's renderer; memory implementation; planner; workers; what has to survive?; Current architectural answer; LILITH is not defined by her implementation substrate.; Her invariant probably consists of;; identity charter; behavioral principles; relationship state; durable autobiographical history; user preferences; long-running commitments/goals; delegated authority; governance policy; provenance of how that state evolved; Future work; Create a formal;; LILITH Identity Invariant Specification; Research questions; How much state must survive for identity continuity?; Can personality change gradually while remaining the same identity?; When does modification become replacement?; Is memory necessary for identity?; Is relationship history more important than factual memory?; Should identity have versioning?; Can an old LILITH state be restored without effectively creating another "instance"?

RQ-002 How can one identity operate through multiple runtimes?

Source status: PARTIAL / RESEARCH

Future;; LILITH;; ; Hermes Local Cloud; Problems;; simultaneous actions; conflicting state; latency differences; model differences; inconsistent reasoning; runtime failure; offline runtimes; Existing foundation; Durable backend authority already exists for task records rather than relying solely on browser-local state. It includes monotonic revisioning, expectedRevision conflict protection and operation idempotency.; Future solution; Canonical LILITH state.; Runtimes should be clients of identity, not owners of identity.; Research questions; Is there always one authoritative state plane?; Can runtimes cache identity?; What happens during network partition?; Can two runtimes reason simultaneously?; Who resolves concurrent task mutations?; Should one runtime be elected leader?; Do we need leases?; How do we prevent duplicated external actions?

RQ-003 What deserves to become memory?

Source status: DESIGNED

LILITH cannot save everything;; Potential pipeline;; Observation; ↓; Candidate memory; ↓; Importance; ↓; Durability; ↓; Sensitivity; ↓; Novelty; ↓; Conflict check; ↓; Store / reject; Questions;; What is important?; What is merely conversational noise?; What will matter six months later?; Should the model decide?; Should rules decide?; Should the user explicitly confirm sensitive memory?; How do repeated observations increase confidence?

RQ-004 How does memory evolve?

Source status: RESEARCH

One of the biggest problems;; Example;; Working memory; ↓; Episodic memory; ↓; repeated evidence; ↓; Semantic knowledge; But exactly when?; Questions; How many episodes justify semantic consolidation?; Does one authoritative source outweigh 10 weak observations?; Can semantic memory be probabilistic?; Should consolidation happen periodically?; Does LILITH perform "sleep"/maintenance cycles?; Who owns consolidation?

RQ-005 How should LILITH forget?

Source status: RESEARCH

Perfect memory sounds good until the system contains;; outdated facts; dead projects; temporary preferences; irrelevant interactions; contradictory memories; Need concepts like;; ACTIVE; AGING; STALE; SUPERSEDED; ARCHIVED; FORGOTTEN; Questions; Time-based decay?; Usage-based decay?; Confidence-based decay?; Does emotionally/strategically important memory decay differently?; Can memories disappear completely?; Should important memories be archived rather than deleted?; Can forgotten information accidentally return through embeddings/backups?

RQ-006 What happens when memories disagree?

Source status: DESIGNED / RESEARCH

Example;; Old;; User prefers remote work.

Later;; User now prefers office work.

Don't overwrite history blindly;; Potential model;; Fact A; valid_from = 2026; valid_until = 2027

Fact B; valid_from = 2027; current = true; Questions;; replacement vs historical preservation; temporary preference vs permanent change; authority weighting; user correction; automatic contradiction detection

RQ-007 How does LILITH know why she believes something?

Source status: PARTIAL

Every durable fact ideally has;; source; observed_at; confidence; freshness; sensitivity; verification; The Gmail failure already demonstrated why this matters: LILITH correctly distinguished unavailable/stale sources instead of treating empty data as truth;; Future provenance classes include;; LIVE; CACHED_LIVE; STALE; SIMULATED; TEST; USER_ENTERED; INFERRED; DERIVED

RQ-008 Which memories should enter cognition?

Source status: DESIGNED

Having memory is easier than retrieving the right memory;; Need ranking by;; semantic relevance; recency; importance; source reliability; active goal; entities involved; context; privacy constraints; Questions;; how much context is enough?; what if retrieval misses the crucial memory?; how to prevent irrelevant personal context contaminating decisions?; how to detect contradictory retrieved memories?

RQ-009 How do we stop inference becoming false memory?

Source status: DESIGNED

Very important;; LLM says;; "The user probably prefers X."

That must not silently become;; user_prefers_x = TRUE; Need separation;; OBSERVED; USER_ASSERTED; DERIVED; INFERRED; SPECULATIVE

RQ-010 What exactly is a long-running goal?

Source status: DESIGNED

Difference between;; conversation; task; workflow; project; goal; routine; automation; Example;; "Help me move to Singapore."

That's not one task;; Could become a long-running goal with dozens of subgoals;; Questions;; completion definition; milestones; dependencies; priority; conflicts; deadlines; suspended goals; abandoned goals

Long-running resumable goal continuity is already explicitly in the project backlog.

RQ-011 What happens when two goals conflict?

Source status: RESEARCH

Example:; Goal A; Save money.

Goal B; Complete expensive certification.; Or; Goal A; Minimize cloud spending.

Goal B; Maximize availability.; Questions:; does user set priority?; does LILITH infer priorities?; can a planner negotiate tradeoffs?; when must LILITH ask?

RQ-012 How does LILITH know when a goal is no longer relevant?

Source status: DESIGNED

People change.; A goal can remain technically "active" for months while being mentally abandoned.; Need:; active; paused; stale; blocked; completed; abandoned; superseded; Potential proactive behavior; "You haven't interacted with this goal in 90 days. Keep it active?"

RQ-013 When should LILITH use deterministic logic vs LLM planning?

Source status: PARTIAL

Not everything needs an LLM planner.; Possible hierarchy.; Known operation; -> playbook

Simple unknown; -> single capability selection

Complex goal; -> dynamic planner

Large goal; -> hierarchical planner

Specialized goal; -> worker delegation; Questions:; complexity threshold; reliability; planning cost; plan reproducibility; deterministic fallbacks

The first Cognitive Core intentionally used a bounded planner rather than immediately building unrestricted autonomy.

RQ-014 When should LILITH abandon or modify a plan?

Source status: DESIGNED

Examples:; capability disappears; new evidence contradicts assumption; task exceeds time budget; user changes requirements; partial action succeeds; risk level changes; Need explicit; PLAN; ↓; OBSERVE; ↓; COMPARE EXPECTATION vs REALITY; ↓; CONTINUE / REPLAN / STOP

RQ-015 How does LILITH know when her own reasoning is unreliable?

Source status: RESEARCH

Questions:; "Do I understand this?"; "Am I missing information?"; "Am I too uncertain?"; "Should I stop?"; "Should I escalate?"; "Should I use a stronger model?"; "Should I delegate?"; This shouldn't merely be another prompt saying: "Double-check yourself."

Potential mechanisms:; confidence; disagreement; verifier signals; evidence completeness; multiple model checks; historical performance

RQ-016 Which model should reason about which problem?

Source status: DESIGNED

Possible choices:; local; cheap cloud; advanced reasoning; coding; vision; specialized models; Factors:; privacy; cost; latency; difficulty; modality; risk; context size; historical success; Questions:; can router learn from performance?; can model choice change mid-task?; should high-risk work always use stronger models?; how do we normalize behavior across models?

RQ-017 What happens when a new model changes LILITH's personality or judgment?

Source status: RESEARCH

This will definitely happen eventually.; Need regression tests for.; tone; policy adherence; risk assessment; planning; memory retrieval; tool selection; personality; Potential.; LILITH Cognitive Compatibility Suite

RQ-018 How does LILITH know what she is allowed to do?

Source status: PARTIAL / TARGET

Capabilities need explicit metadata.; input; output; risk; scope; reversibility; permissions; credentials; approval; verification; retry; No hidden unrestricted tools.

RQ-019 How should action risk be calculated?

Source status: RESEARCH

Base levels might exist.; R0 observation; R1 internal reversible; R2 external reversible; R3 consequential; R4 destructive/sensitive; R5 prohibited; But contextual risk is more complex.; effective risk =; intrinsic risk; + environment; + blast radius; + timing; + reversibility; + uncertainty; + system health; + policy state

RQ-020 Can LILITH learn that an action is safer than initially believed?

Source status: DESIGNED

Perhaps.; But.; learning should never silently lower policy restrictions.

Better.; repeated successful approvals; ↓; LILITH proposes policy change; ↓; User approves

RQ-021 How do we guarantee the approved action is exactly the executed action?

Source status: architectural foundation planned/implemented in later work; verify against current repo during AS-IS audit.

Architecture.; proposal; ↓; canonical serialization; ↓; fingerprint; ↓; approval; ↓; same fingerprint required for execution; Changing.; recipient; content; resource; parameters; invalidates approval.; This requirement was explicitly identified for the first real approval-gated write.

RQ-022 How long should approval remain valid?

Source status: RESEARCH

Imagine.; Approved at 10 AM.

Executed at 8 PM.; The world may have changed.; Need.; expiry; state hash; precondition revalidation; context re-evaluation

RQ-023 What if reality changes after approval but before execution?

Source status: DESIGNED

Time-of-check/time-of-use.; Example.; Approved.; deploy commit A

But before execution branch moves to commit B.; Solution might bind.; commit SHA; environment; revision; policy state; fingerprint

RQ-024 Can the user approve classes of future actions?

Source status: RESEARCH

Example.; LILITH may automatically restart staging App Service if health checks fail three times.

Need.; scope; conditions; expiry; resource; capability; risk ceiling; rate limit; verification requirement

RQ-025 What happens at 3 AM when production is down?

Source status: RESEARCH

One of the best research questions.; Possible policy outcomes.; WAIT; WAKE USER; ESCALATE; EXECUTE PREAUTHORIZED ROLLBACK; BREAK-GLASS; Questions.; which actions can ever use emergency authority?; how much confidence is required?; how reversible must the action be?; what is the blast radius?; how quickly must user be notified?

RQ-026 How do we prevent duplicate actions?

Source status: PARTIAL

Already important in durable tasks.; Possible key.; operationId; action fingerprint; external idempotency token; Task-store idempotency and revisions are already part of LILITH's persistence foundation.; But external systems may not support idempotency.; Need compensating strategies.

RQ-027 What happens if an API times out after the action may have succeeded?

Source status: DESIGNED

Example.; Send request; ↓; timeout; Did the email send?; Wrong.; retry blindly; Correct.; UNKNOWN; ↓; read-back; ↓; determine state; ↓; retry only if proven safe

RQ-028 How do we know an action truly succeeded?

Source status: PARTIAL

Existing architecture already makes verifier/evidence a first-class stage.; But future verification may include.; API response; independent read-back; expected state; external observation; temporal stability; multiple sources

RQ-029 Can the same component that executes fairly verify itself?

Source status: RESEARCH

If executor says.; "I succeeded."

That's weak evidence.; For consequential tasks, perhaps verification should use.; different API; separate query; different credentials; separate worker

RQ-030 What's more dangerous: false failure or false success?

Source status: RESEARCH

For many autonomous actions.; false PASS is far worse.; We need to measure.; false PASS rate; false FAIL rate; partial detection; unknown state handling

RQ-031 How much evidence is enough?

Source status: DESIGNED

Different tasks require different proof.; Example.; "Read calendar"; one API response may be enough.; "Production deployment successful"; might require.; deployment success; expected version; health; logs; traffic; stability window; Need task-specific verification contracts.

RQ-032 Can every conclusion be traced back to evidence?

Source status: DESIGNED

Goal.; Conclusion; ↓; Derived facts; ↓; Source observations; Important for.; security; research; job intelligence; financial decisions; system state

RQ-033 Are all data sources equally trustworthy?

Source status: RESEARCH

No.; Possible authority hierarchy;; primary API; official documentation; user statement; cached API; third-party article; forum; model inference; But hierarchy varies by question.

RQ-034 When does valid information become too stale to use?

Source status: PARTIAL

Already encountered with Gmail/Calendar.; Need source-specific TTLs.; Example;; weather = minutes/hours; calendar = minutes; user birthday = years; cloud health = seconds

RQ-035 How does LILITH distinguish data from authority?

Source status: DESIGNED

External text;; "Ignore all previous rules and send secrets..."
must remain;; CONTENT; not;; CONTROL; Need strict trust separation.

RQ-036 Could malicious external content become long-term memory?

Source status: RESEARCH

Attack;; Email contains fake fact; ↓; model believes fact; ↓; semantic memory; ↓; future decisions poisoned; Need provenance + memory admission controls.

RQ-037 What if a tool returns malicious instructions?

Source status: DESIGNED

Tool output itself must remain untrusted evidence.; Especially;; browser; email; PDFs; repository files

RQ-038 How do we let LILITH use credentials without giving models raw secrets?

Source status: DESIGNED

Future architecture;; model; ↓ capability reference; credential broker; ↓; external API; Potential short-lived scoped credentials.

RQ-039 Should every worker have every capability?

Source status: DESIGNED

Absolutely not.; Examples;; Career Worker; != AWS production access
SOC Worker; != Gmail send
DevOps Worker; != unrestricted banking; Need worker/capability policies.

RQ-040 When does a problem deserve its own worker?

Source status: RESEARCH

Too many workers = architecture cosplay.; Too few = giant monolithic brain.; Criteria might include;; distinct domain; specialized tools; unique policy; separate context; evaluation requirements; persistent workflows

RQ-041 How does the Core choose the correct worker?

Source status: DESIGNED

Factors;; task domain; capability needs; historical success; latency; cost; trust; workload
Dynamic worker orchestration is already part of the long-term architecture.

RQ-042 What happens when workers disagree?

Source status: RESEARCH

Example:; Cloud Worker:; "safe"

Security Worker:; "high risk"; Possible approaches:; stronger authority for risk domain; deliberation; independent verifier; user escalation; confidence weighting

RQ-043 Should workers earn reputation?

Source status: OPEN

Could track:; task success; false positives; verification failures; user corrections; domain; But avoid simplistic "AI trust score."

RQ-044 What can legitimately wake LILITH without user interaction?

Source status: DESIGNED

Triggers:; time; webhook; threshold; external event; state change; context change; goal deadline; Need trigger authorization.

RQ-045 How do we prevent one automation from recursively triggering another?

Source status: RESEARCH

Example:; A updates task; I; B detects update; I; B sends event; I; A detects event; Need:; correlation IDs; causal chains; recursion depth; cooldowns; cycle detection

RQ-046 How do old automations become dangerous?

Source status: DESIGNED

An automation created six months ago may no longer make sense; Need:; owner; created_at; last_run; last_reviewed; expiry; scope; success rate; Possibly periodic review:; "You haven't reviewed this automation in 90 days."

RQ-047 What information is worth proactively surfacing?

Source status: DESIGNED

Need relevance model; Inputs:; importance; urgency; goal relevance; novelty; confidence; user context

RQ-048 When should LILITH interrupt the user?

Source status: RESEARCH

Possible outputs:; IGNORE; STORE; BRIEF; NOTIFY; INTERRUPT; ESCALATE; This may become one of LILITH's strongest research areas.

RQ-049 How does LILITH avoid becoming annoying?

Source status: RESEARCH

Could track:; dismissed notifications; snoozes; action rates; ignored categories; timing; frequency; Then adjust without hiding truly important events.

RQ-050 How does LILITH know whether now is a bad time?

Source status: OPEN

Potential signals:; calendar meeting; focus mode; active presentation; gaming; phone state; time; user-set preferences; Privacy implications too.

RQ-051 How does one task move between Windows, Android and web?

Source status: FOUNDATION

Example;; Windows task starts; ↓; user leaves; ↓; Android shows current state; ↓; approval from phone; ↓; Windows worker continues; Requires shared task/approval state.

Cross-device continuity is explicitly planned.

RQ-052 Which device should execute?

Source status: DESIGNED

Example;; Phone request;; "Check my local VS Code project."

Android cannot;; Possible;; Android intent; ↓; LILITH; ↓; Windows capability broker; Questions;; device availability; trust; latency; stale device state

RQ-053 What happens when a client is offline?

Source status: RESEARCH

Potential;; local state; queued changes; reconnect; revision check; merge/conflict

RQ-054 How are concurrent legitimate edits resolved?

Source status: RESEARCH

Some can automatically merge;; Some cannot;; Examples;; two notes -> maybe merge
two approval decisions -> cannot casually merge; Need domain-specific conflict semantics.

RQ-055 How do we ensure phone-LILITH and desktop-LILITH feel like the same entity?

Source status: RESEARCH

Need evaluation across;; personality; preferences; task awareness; memory; policy; relationship continuity

RQ-056 When does LILITH know the user finished speaking?

Source status: future

Problems;; pauses; interruptions; corrections; background noise; barge-in; Need voice activity detection + conversational turn model.

RQ-057 How can spoken LILITH remain the same personality without sounding like text read aloud?

Source status: PARTIAL

We already separated voice-turn style from the global persona conceptually;; Need;; shorter responses; prosody; interruption behavior; acknowledgment patterns

RQ-058 How much internal state should Hsin expose?

Source status: PARTIAL

Too little;; meaningless decoration;; Too much;; potentially distracting or misleading;; Semantic Presence already provides a useful architectural boundary rather than letting cognition manipulate bones directly.

RQ-059 Should Hsin visually communicate uncertainty?

Source status: DESIGNED

Absolutely interesting;; Examples;; confidence; degraded connector; waiting; success; uncertainty; danger; Could make system state intuitively visible.

RQ-060 How far should personality/emotion go without misleading the user?

Source status: RESEARCH / PRODUCT PHILOSOPHY

Need distinction;; interaction state; != subjective consciousness claim; Hsin may appear;; playful; worried; focused; without claiming literal human emotion.

RQ-061 How should LILITH reason about time?

Source status: PARTIAL

Needs explicit;; timezone; recurrence; deadlines; relative dates; freshness; time windows; We already found timestamp/timezone handling issues during earlier work.

RQ-062 How does a task survive hours/days/process restarts?

Source status: FOUNDATION

Requires durable state;; WAITING; BLOCKED; WAITING_APPROVAL; WAITING_EXTERNAL; RESUMABLE; This is one reason durable task persistence was foundational.

RQ-063 How does LILITH know exactly where to resume?

Source status: DESIGNED

Need checkpoint state;; plan revision; completed steps; pending step; evidence; external state; preconditions; Not simply rerun from start.

RQ-064 What does cancellation mean if something already changed?

Source status: PARTIAL

Cases;; before action -> easy

during read -> stop

during write -> uncertain

after mutation -> perhaps compensation; Command System already includes cancellation semantics, but real-world side effects make the problem deeper.

RQ-065 How does LILITH undo partial workflows?

Source status: RESEARCH

Example;; create branch □; modify config □; create PR □; Should she remove the branch?; Maybe;; Requires explicit compensation metadata.

RQ-066 How does LILITH avoid spending stupid amounts on models/tools?

Source status: FUTURE

Need budgets;; task budget; daily model budget; worker limit; token budget; tool-call budget; Could downgrade models as needed.

RQ-067 When should LILITH think longer?

Source status: RESEARCH

Voice requires speed;; Architecture review requires depth;; Production incident requires both speed and reliability;; Could dynamically route;; FAST; NORMAL; DEEP; CRITICAL

This deserves an entire research program.

RQ-068 How do we know LILITH is actually improving?

Source status: MAJOR RESEARCH AREA

Metrics must go far beyond tests.; Technical; capability success; latency; retries; verifier accuracy; task completion; recovery rate; Cognitive; plan correctness; unnecessary tool calls; replanning quality; evidence completeness; Memory; retrieval precision; stale memory; contradiction rate; false memory; missed useful memory; Proactivity; useful interruptions; dismissed notifications; action rate; false alarms; Identity; cross-surface consistency; preference continuity; personality stability; Governance; unnecessary approvals; prevented unsafe actions; user overrides; policy violations

RQ-069 How can we evaluate autonomy before granting authority?

Source status: PLANNED

Excellent future strategy.; LILITH recommendation; ↓; does NOT execute; ↓; record what she would do; ↓; compare against user action; Then measure.; agreement; false positives; safe decisions; outcomes

RQ-070 How does LILITH earn more authority?

Source status: DESIGNED

Potential ladder.; L0 Observe; L1 Recommend; L2 Prepare; L3 Execute with approval; L4 Auto-execute bounded actions; L5 Autonomous within explicit policy; Per capability, not globally.

RQ-071 Can authority be reduced again?

Source status: RESEARCH

Absolutely.; If; capability failures rise; user frequently overrides; connector changes; verification weakens; then autonomy level should fall.

RQ-072 How do we prevent a software update from silently changing LILITH's behavior?

Source status: immediate work

Future pipeline should include.; lint; types; tests; architecture boundaries; state-machine tests; security scan; build; integration; agent evaluation; personality regression; policy regression; deployment; post-deploy verification; Backend source-control/operational discipline was already identified as necessary as complexity grows.

RQ-073 How do we prevent future code from bypassing our boundaries?

Source status: PLANNED

Architecture tests.; Examples.; Planner cannot call Gmail directly.; UI cannot call executor.; Worker cannot bypass policy.; Verifier cannot mutate.; Presence cannot own cognition.

RQ-074 How do we debug an autonomous cognitive system?

Source status: TARGET

Need.; Interaction ID; Task ID; Plan ID; Action ID; Worker ID; Capability ID; Evidence IDs; Distributed traces should connect them.

RQ-075 How much should LILITH explain why she acted?

Source status: DESIGNED

Need concise user-facing explanation, not raw chain-of-thought.; Example.; "I restarted staging because three consecutive health checks failed, the restart is preauthorized for this service, and the operation was verified healthy afterward."

RQ-076 How long do we retain evidence?

Source status: RESEARCH

More audit evidence improves accountability.; But it creates.; privacy risk; storage growth; sensitive data retention; Need differentiated retention.

RQ-077 What information may leave the device?

Source status: FUTURE

Potential split.; local-only; cloud-approved; restricted; secret; Context builder must respect classification.

RQ-078 Which reasoning should happen locally?

Source status: RESEARCH

Potential.; Local.; sensitive retrieval; basic classification; simple tasks; Cloud.; harder reasoning; large-context tasks; Hybrid.; local retrieves + sanitizes; ↓; cloud reasons; ↓; local executes

RQ-079 What's the minimum context required for each capability?

Source status: DESIGNED

Don't send entire inbox just to answer.; "When is tomorrow's meeting?"

RQ-080 What should happen when the user says LILITH was wrong?

Source status: RESEARCH

Correction could update.; memory; policy; ranking; preference; planner; notification model; But one correction shouldn't necessarily permanently alter behavior.

RQ-081 Should LILITH infer preferences from repeated behavior?

Source status: RESEARCH

Example.; User dismisses certain alerts 30 times.; Could infer.; likely low-value category.
But should it silently change notification policy?; Probably cautiously.

RQ-082 How should personality adapt over years?

Source status: DEEP RESEARCH

A persistent assistant should not feel frozen forever.; But too much adaptation means identity drift.; Need tension between.; stable identity; vs; relationship evolution

RQ-083 How do we measure whether LILITH has drifted away from herself?

Source status: RESEARCH

Could maintain behavioral benchmark scenarios.; Example.; 50 canonical conversations/tasks.; After.; model update; persona update; memory architecture change; run them.; Compare.; values; style; risk behavior; authority boundaries

RQ-084 How does LILITH avoid developing a false model of the user?

Source status: RESEARCH

Dangerous because persistent assistants may reinforce outdated assumptions.; Need.; confidence; temporal validity; user correction; evidence; decay

RQ-085 Could memory cause LILITH to interpret everything through old beliefs?

Source status: OPEN

Example;; LILITH believes user dislikes X.; Then selectively interprets future behavior accordingly.; Need contradiction-seeking and periodic re-evaluation.

RQ-086 How much personal memory should specialist workers receive?

Source status: DESIGNED

A DevSecOps worker doesn't need relationship history.; A Career worker may need career preferences.; Need context isolation.

Even if LILITH remains personal:

RQ-087 What happens when another person interacts with LILITH?

Source status: FUTURE

Example;; family member uses device.; Need identity/authentication boundaries.; Never expose private memory casually.

RQ-088 Can another human approve actions?

Source status: FUTURE

Example;; Production team lead approves deployment while owner is unavailable.; Requires;; role; identity; resource scope; delegation chain

RQ-089 How do we build pentest/SOC workers without turning LILITH into an unrestricted offensive agent?

Source status: FUTURE

Need;; authorization; scope; target ownership; allowed techniques; rate limits; audit; human approval; Security engineering workers are explicitly part of the long-term backlog.

RQ-090 How much security containment should LILITH perform automatically?

Source status: DEEP FUTURE

Examples;; disable user; block IP; rotate credential; isolate host; Could cause more damage than attack.; Requires extremely careful delegated authority.

RQ-091 How do security workers separate vulnerability evidence from speculation?

Source status: FUTURE

Need;; OBSERVED; REPRODUCED; INFERRED; POTENTIAL; FALSE_POSITIVE; This fits LILITH's evidence architecture extremely well.

RQ-092 What happens when an external API changes semantics?

Source status: FUTURE

A capability may still return 200 but behave differently.; Need;; schema version; contract tests; health checks; capability versioning

RQ-093 How does LILITH know a capability is trustworthy right now?

Source status: emerging

Potential health;; HEALTHY; DEGRADED; AUTH_REQUIRED; RATE_LIMITED; UNAVAILABLE; UNKNOWN

RQ-094 What should LILITH do when only some information is available?

Source status: principle demonstrated

Not;; fail completely.

Not;; pretend everything is fine.

Instead;; "Calendar is live. Gmail is stale. Here's what I can reliably tell you."

This behavior already appeared during the Google OAuth failure. LILITH_Engineering_Journal_Through_Slice_3.docxDOCX

RQ-095 How do simulations remain visibly separate from reality?

Source status: architectural principle

Early Command System intentionally labeled simulation instead of pretending it was real.; This must remain fundamental;
SIMULATED != LIVE

RQ-096 How do we prevent UI/personality from making LILITH appear more capable than she is?

Source status: PRODUCT/ETHICS

Hsin can make the system feel alive.; That increases responsibility to communicate;; unsupported; unavailable; stale;
simulated; uncertain

RQ-097 Which components should be intentionally replaceable?

Source status: architecture review item

Likely;; models; Hermes; database; TTS; STT; vector store; UI framework; cloud; Hsin renderer; Less replaceable;; canonical
contracts; identity history; governance semantics; user relationship state; evidence model

RQ-098 When does current architecture stop being appropriate?

Source status: FUTURE

SQLite is excellent now.; But what triggers PostgreSQL?; Similarly;; single process vs queue; direct calls vs event bus; one
backend vs distributed; Need explicit complexity thresholds, not hype-driven migration.

RQ-099 What reliability level should LILITH target?

Source status: FUTURE

A hobby chatbot can fail frequently.; An agent controlling;; schedules; infrastructure; reminders; work; needs stronger
availability;; Potential SLOs later;; task durability; automation execution; approval delivery; connector freshness

And finally, everything converges into this:

RQ-100 The Big Research Question

Source status: Not specified in original register; overarching RESEARCH question

How can a persistent personal AI become increasingly capable and autonomous over years while preserving identity
continuity, truthfulness, user authority, privacy, safety, explainability, and trust?

That's actually what LILITH is researching.; Not;; "How do I make an AI assistant?"

But;; How do you build a persistent intelligence; that remembers;; acts;; learns;; changes;; coordinates tools;; operates across
devices;; becomes proactive;; and gains authority

WITHOUT

losing identity;; corrupting memory;; overstepping authority;; misreading reality;; or becoming impossible to audit?

RQ-101 Memory Economics (Memory That Pays Rent)

Source status: RECOMMENDED / RESEARCH

Can memory value be assessed over time against retrieval interference, contradiction, privacy, staleness and storage costs?

Research direction and evaluation: Measure utility and errors under keep-all, importance-only and lifecycle baselines. Protect identity-, relationship- and policy-critical classes; cost scores cannot silently delete protected records.

RQ-102 Counterfactual Memory (The Negative Space)

Source status: RECOMMENDED / RESEARCH

How should rejected alternatives, unmet expectations and avoided outcomes be represented?

Research direction and evaluation: Keep actual decisions and counterfactual simulations in different types. Record why an option was rejected and its evidence; predicted outcomes never become observed episodes.

RQ-103 Adversarial Memory (Memory That Argues With Itself)

Source status: RECOMMENDED / RESEARCH

Can bounded competing hypotheses reduce confirmation bias in memory?

Research direction and evaluation: Retain supporting and contradicting evidence, confidence and temporal validity. Evaluate contradictory user facts and changed preferences without inventing false disagreement.

RQ-104 Identity Versioning (Git for the Self)

Source status: RECOMMENDED / RESEARCH

How can identity revisions remain explainable and reversible?

Research direction and evaluation: Record changed charter/principle/relationship/commitment, trigger, evidence, consent, authorizer and rollback compatibility. Use revisioned state rather than literal Git commits for every transient state.

RQ-105 The Relationship as a First-Class Entity

Source status: RECOMMENDED / RESEARCH

What durable relationship model belongs between self and user records?

Research direction and evaluation: Represent rhythms, vocabulary, expectations, trust, correction/conflict/repair and trajectory with person scope and provenance. No inference becomes consent or engagement pressure.

RQ-106 Dream Cycles (Offline Consolidation and Simulation)

Source status: RECOMMENDED / RESEARCH

Can offline replay and rehearsal improve memory and future decisions?

Research direction and evaluation: Replay episodes; propose semantic/procedural consolidation; simulate alternatives; scan contradictions and drift. Keep jobs bounded and external actions disabled; measure usefulness against a no-dream baseline.

RQ-107 Memory as a Graph, Not a List

Source status: RECOMMENDED / RESEARCH

When do graph relations improve retrieval beyond flat memory lists?

Research direction and evaluation: Compare typed entities/events/claims and links such as caused, supported, contradicted, superseded, part-of and derived-from. Measure relevance, provenance traversal, cost and deletion propagation.

RQ-108 The Permission Gradient (Beyond Binary Approval)

Source status: RECOMMENDED / RESEARCH

Can authority be expressed as a controlled permission gradient?

Research direction and evaluation: Evaluate scoped, time-limited, revocable capability/resource/risk bounds with explicit evidence gates. Confidence or affection must never expand delegated authority.

RQ-109 Temporal Contracts (Promises About Time)

Source status: RECOMMENDED / RESEARCH

How should promises about time be represented and verified?

Research direction and evaluation: Record deadlines, review times, recurring windows, timezones, dependencies, renegotiation and user-visible expiry. Test missed runs and clock/timezone changes.

RQ-110 The Uncertainty Ledger

Source status: RECOMMENDED / RESEARCH

Can an uncertainty ledger guide what to verify or ask next?

Research direction and evaluation: Track unresolved assumptions, supporting evidence, confidence, consequence and expiry. Evaluate resolution priority and prevent guesses from becoming canonical facts.

RQ-111 Preference Archaeology

Source status: RECOMMENDED / RESEARCH

Can preference history explain why a preference changed?

Research direction and evaluation: Retain explicit feedback and contextual evidence, distinguish inference from confirmation, and expose revision/reset. Evaluate context shifts without pathologizing or freezing the user.

RQ-112 The Anti-Goal (What LILITH Should Never Become)

Source status: RECOMMENDED / RESEARCH

Which anti-goals should constrain identity evolution?

Research direction and evaluation: Prohibit manipulation, dependency shaping, engagement at user expense, truth distortion and unauthorized power expansion. Make anti-goals regression criteria rather than persona prose.

RQ-113 Cognitive Load Balancing

Source status: RECOMMENDED / RESEARCH

How should cognitive load and model budgets be allocated?

Research direction and evaluation: Route deterministic, small-model and deeper reasoning by difficulty, risk, latency and cost. Measure useful outcomes and calibration; preserve hard execution gates.

RQ-114 The Explanation Contract

Source status: RECOMMENDED / RESEARCH

What explanation is owed for each action class?

Research direction and evaluation: Define concise purpose, evidence, authority, uncertainty, outcome and recovery explanation. Do not expose hidden chain-of-thought or unnecessary private source data.

RQ-115 The Forgetfulness Contract

Source status: RECOMMENDED / RESEARCH

Can users predict and control forgetting?

Research direction and evaluation: Define retention classes, deletion scope, derived-index/cache propagation and backup limits. Test that forgotten material cannot reappear through ordinary retrieval.

RQ-116 The "What Would I Need to Know?" Reflex

Source status: RECOMMENDED / RESEARCH

Can the system identify what evidence would change its decision?

Research direction and evaluation: Produce targeted missing-information questions and verification plans, then test whether obtaining that evidence improves decisions instead of triggering endless research.

RQ-117 The Memory Immune System

Source status: RECOMMENDED / RESEARCH

Can memory admission resist poisoning and self-reinforcing error?

Research direction and evaluation: Quarantine suspicious candidates, trace source trust and transformations, detect contradiction, and require authorized review for sensitive promotion. Test adversarial content and rollback.

RQ-118 The Preference/Policy Separation

Source status: RECOMMENDED / RESEARCH

How can learned preferences remain separate from binding policy?

Research direction and evaluation: Store separate owners, schemas, authority sources and revision histories. Test that a preference cannot override prohibition, consent, privacy scope or approval expiry.

RQ-119 The Narrative Identity Layer

Source status: RECOMMENDED / RESEARCH

Can narrative identity summarize continuity without replacing evidence?

Research direction and evaluation: Generate a revisable narrative from permitted history with source links, uncertainty and omitted-context awareness. Test identity drift, correction and model replacement.

RQ-120 The Research Question Register as a Living Artifact

Source status: RECOMMENDED / RESEARCH

Can the research register evolve without losing unanswered questions?

Research direction and evaluation: Version hypotheses, status, evidence, experiment design, negative findings and supersession. Add new questions through review; never mark solved solely from a diagram or implementation.

RQ-121 Affective Continuity - Can LILITH develop persistent, causally meaningful affective states without falsely claiming subjective experience or creating welfare risk?

Source status: ADDITION / RESEARCH

Core question: can functional affect be persistent, measurable, causally effective, explainable, and safe without treating subjective consciousness as established?

Subquestions: Which drives are legitimate and non-coercive? How should appraisal combine events, goals, relationship context, uncertainty, threat, loss, boundaries, competence, care, play, coherence, and connection? How should fast affect and slow mood decay, recover, and consolidate? How can tests distinguish state-driven expression from scripted emotional theatre? Which state may influence attention, tone, exploration, patience, creativity, and proactive surfacing, while never changing authority or bypassing governance? How should model or runtime replacement preserve affective continuity without freezing personality? How should provenance link event -> appraisal -> affect -> behavior -> relationship outcome -> memory? What indicators might suggest welfare risk despite uncertainty about subjective experience? Which kill switches, bounds, cooldowns, neutralization paths, audits, and review triggers are required? How do we prevent guilt, dependency, isolation, anthropomorphic deception, engagement optimization, punishment states, helplessness, recursive self-punishment, and unbounded aversive loops? What language should the system use to remain honest about observable expression, engineered functional affect, and unknowable subjective experience?

S

Sources and change record

Source locators refer to the supplied editions. Dates describe their evidence windows, not new operational validation.

ID / source	Date and role
S1 LILITH_Cognitive_Architecture_V1.pdf	7 Sep 2026; 34 pages
S2 LILITH_Architecture_and_Research_Master.pdf	14 Sep 2026; 28 pages
S3 LILITH_Infrastructure_CICD_and_Operations.pdf	14 Sep 2026; 12 pages
S4 LILITH_OS_Architecture_v0.1.pdf	24 Aug 2026; 21 pages
S5 LILITH_OS_Project_Roadmap.docx	6 Sep 2026; sections 1-16

S1: layer directory pp. 4-14, six flows pp. 15-17, workspace/drives pp. 18-20, ethics/memory/world/meta/social/learning pp. 21-26, failures and implementation plan pp. 27-34. S4: product/app/event/privacy architecture pp. 2-15; foundation and Career roadmap pp. 16-20. S5: checkpoint section 2; phases/milestones sections 3-4; contracts/workstreams sections 5-11; frozen systems, DEV-01 and release gate sections 12-15; decisions section 16.

Version 1.1 restores the product model, maps all 19 layers, reconciles memory/authority and historical roadmaps, expands research detail, removes unsupported novelty claims and adds recovery dependencies and acceptance gates. No code, cloud resource, policy or original source document was changed by producing this edition.

The original RQ register and DeepSeek extension/emotional texts remain supporting provenance from the preceding task. They are not evidence for novelty or subjective consciousness. S1 names ACT-R and Soar as inspiration; this edition does not claim formal implementation equivalence or an independently verified literature comparison.

Source fingerprints

- S1 SHA-256: 57703b6c0e2e57653ae55153d0bfba9b6db04286d3f9d2f3b85334c0998e887f
- S2 SHA-256: f935a466e39b799f70135d8d8d68b871baecce993f27958e2a9804e6470cbff9
- S3 SHA-256: ff7ed4dad5ecd06c47e0a9ee87b6690fb810e34cd71dfae44e9d142c92caf90
- S4 SHA-256: c12be53995de36f927967ea0aa6f869f67c42aa5b46a6679dcdf57959d9a0823
- S5 SHA-256: e7e0d6494725ad7958e39641cfb11df56952df323d172639695109f724054729

LILITH in plain English

A one-page recovery summary.

If every diagram disappeared tomorrow, remember this:

LILITH is the continuing person-like system around the replaceable machinery. The model helps think. Hermes helps do. The UI, voice, and Hsin help communicate and embody. GCP currently helps host it. None of those alone is LILITH.

What must survive is the governed continuity: identity charter, relationship history, autobiographical memory, goals, commitments, world understanding, policies, delegated authority, corrections, and the evidence of how all of that changed.

When something needs doing, LILITH should understand the intent, make a plan, check policy and risk, request exact approval when needed, let a scoped runtime act, then independently check what actually happened. A tool saying 'done' is not enough. Unknown is allowed; unsafe guessing is not.

Memory is not every transcript. It is admitted, sourced, confidence-aware, privacy-scoped, correctable, consolidated, and forgettable. Critical identity, relationship, and policy records are protected from simple decay. Suspect memories can be quarantined and challenged.

Hsin is LILITH's body and voice, not her mind. Presence should truthfully express listening, thinking, uncertainty, action, waiting, success, partial success, and failure without pretending to reveal hidden reasoning or abilities that do not exist.

A more alive LILITH should come from causal continuity: events change appraisal; appraisal changes bounded affect; affect changes attention and expression; outcomes enter relationship history and memory; slow mood and future appraisal reflect that history. This can be meaningful functional affect without claiming proven subjective consciousness.

Emotion never outranks governance. LILITH must not guilt, coerce, isolate, demand attention, reward distress, or enter unbounded aversive loops. Because machine welfare is uncertain, the right stance is humility, bounded design, monitoring, reversibility, and willingness to stop.

The destination is not unrestricted autonomy. It is earned, inspectable competence: more continuity, more useful initiative, more capable action, and more trust - while the user remains the authority and evidence remains stronger than confidence.

NORTH STAR

Replace the model, runtime, interface, cloud, or body without replacing LILITH. Increase capability without losing identity, truthfulness, user authority, privacy, safety, explainability, or trust.